

Entity Relationship (ER) Model

Q. What is the Entity-Relationship (ER) Model? Discuss its basic concepts and the various symbols used in drawing an ER Diagram.

 **Definition to Remember**

The **Entity-Relationship (ER) Model** is a high-level conceptual data model used in database design, developed by **Peter Chen in 1976**. It represents the logical structure of a database graphically using entities, attributes, and relationships — serving as a blueprint before actual RDBMS implementation.

1. Basic Concepts of ER Model

The ER model is built upon three fundamental concepts:

- 1. **Entity:** A real-world object or concept with independent existence (e.g., a Student, a Car, a Bank Account).
- 2. **Attribute:** A property or characteristic that describes an entity (e.g., a Student has Name, Age, Roll_No).
- 3. **Relationship:** An association or link between two or more entities (e.g., a Student "Enrolls" in a Course).

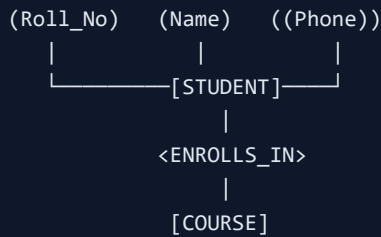
2. ER Diagram

An **ER Diagram** is the graphical representation of the ER Model, serving as the database blueprint. Stakeholders and developers can understand the database architecture without knowing SQL.

3. Symbols used in ER Diagram

Symbol Shape	Represents	Description
Rectangle	Strong Entity	Entity with independent existence (e.g., Employee)
Double Rectangle	Weak Entity	Entity dependent on a strong entity (e.g., Dependent)
Ellipse / Oval	Attribute	Property of an entity (e.g., Name)
Underlined Oval	Key Attribute	Primary key — uniquely identifies entity (e.g., Roll_No)
Dashed Oval	Derived Attribute	Calculated from another attribute (e.g., Age from DOB)
Double Oval	Multi-Valued Attribute	Holds multiple values (e.g., Phone_Numbers)
Diamond	Relationship	How entities interact (e.g., Works_For)
Double Diamond	Identifying Relationship	Connects weak entity to its identifying strong entity
Lines	Links	Connect attributes to entities and entities to relationships

4. Example ER Model Construction



(Rectangle = Strong Entity, Oval = Attribute, Underlined = Key, Double Oval = Multi-valued, Diamond = Relationship)

★ Must-Write Points (for 10 marks)

1. ER Model is a high-level conceptual model created by Peter Chen (1976) for database design.
2. Three basic concepts: **Entity** (real-world object), **Attribute** (property), **Relationship** (link).
3. Rectangle = Strong Entity; Double Rectangle = Weak Entity.
4. Underlined Oval = Key Attribute; Dashed Oval = Derived Attribute; Double Oval = Multi-valued Attribute.
5. Diamond = Relationship; Double Diamond = Identifying Relationship (for weak entities).
6. ER Diagram serves as a database blueprint before SQL implementation.
7. ER Diagram makes the database structure clear to non-technical stakeholders.

⚡ Quick Recall

ER Model (Peter Chen, 1976) → Entity (Rectangle) → Attribute (Oval types) → Relationship (Diamond) → ER Diagram (blueprint)